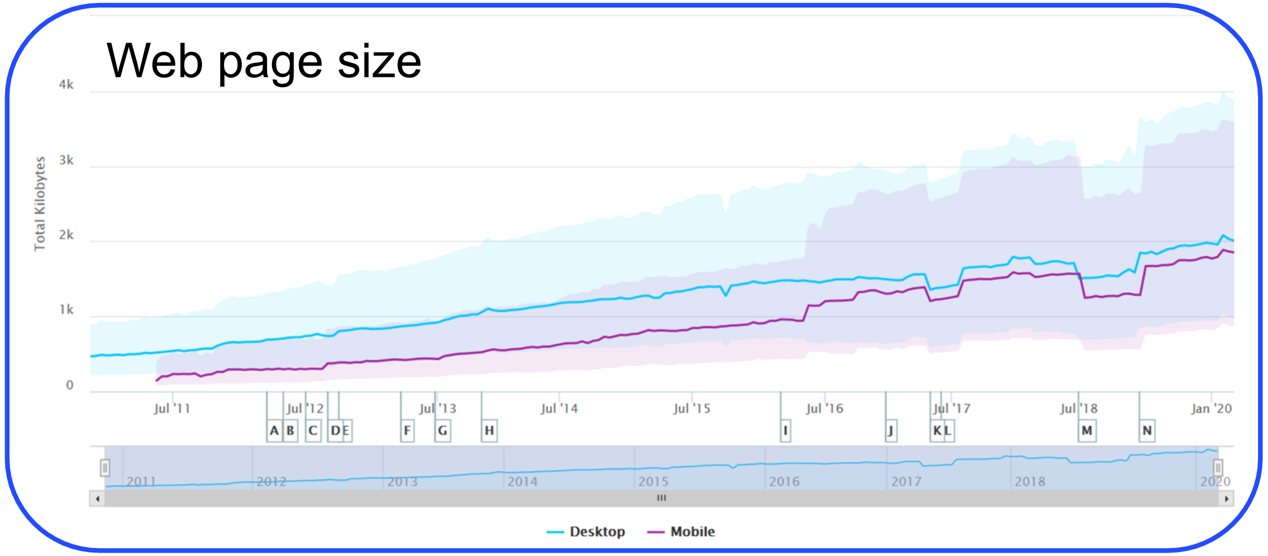
A New Transport Protocol

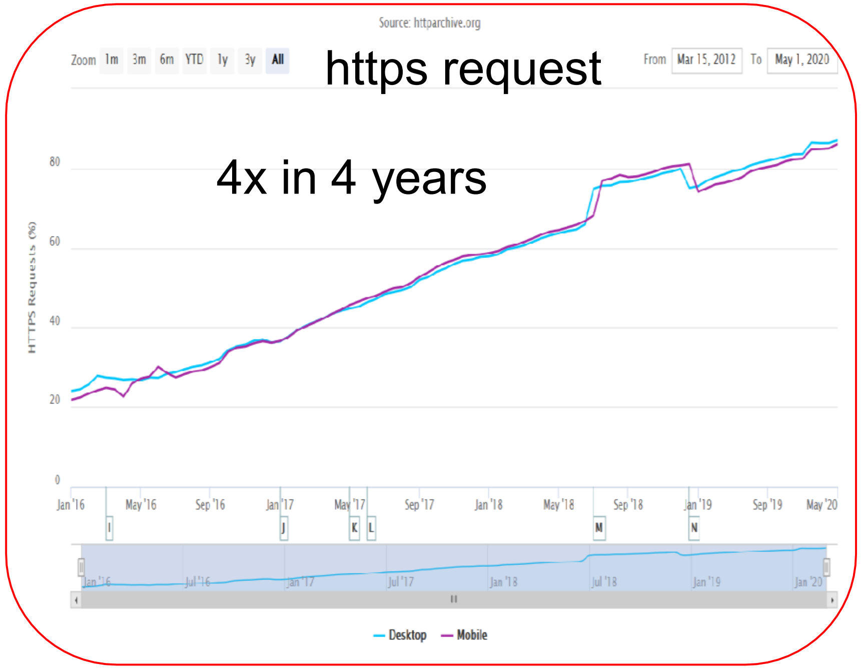
QUIC: Quick UDP Internet Connections

HTTP/3: HTTP over QUIC is next Generation

Introduction: Change

- **Increasing scale of .... everything**
  - Flow size changes
  - Flow count increases (e.g., web pages)
  - Flow diversity increase (e.g., web pages)
    - Multiple connections





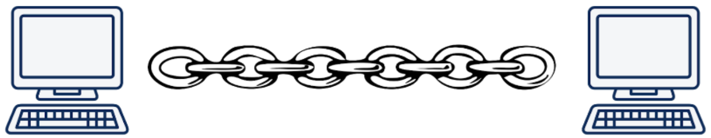
HTTP Network Stack

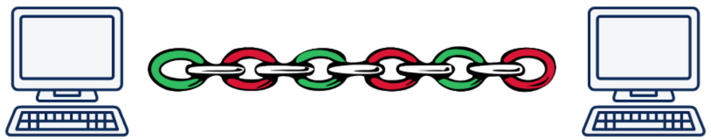
HTTP
TLS 1.2+
TCP
IP

HTTP / 1.1
- January 1997
- **Many parallel TCP connection (6 connections per host name)**
- **HTTP head of line blocking**

HTTP / 2
- May 2015
- **Using Single connection per host**
- **Many parallel streams**
- **TCP head of line blocking**

TLS – Transport Layer Security
TCP – Transport Control Protocol
IP – Internet Protocol

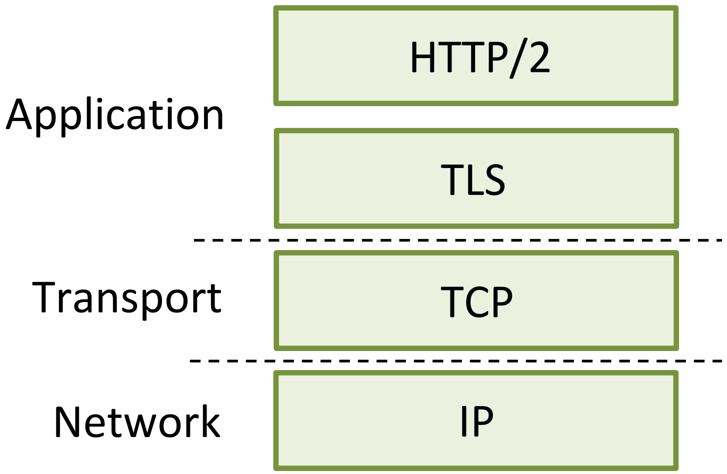




https://http3-explained.haxx.se/en/why-quic

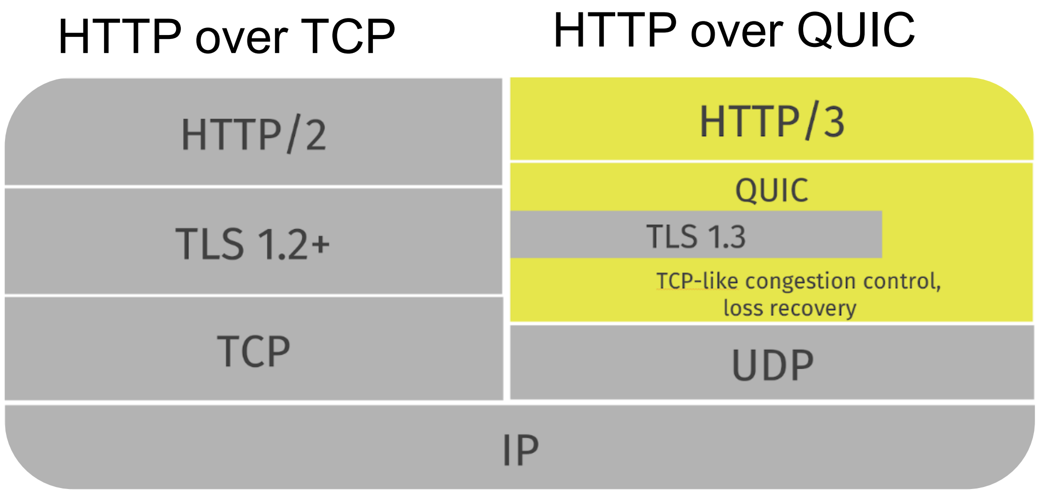
QUIC: Quick UDP Internet Connections

- application-layer protocol, on top of UDP
  - increase performance of HTTP
  - deployed on many Google servers, apps (Chrome, mobile YouTube app)



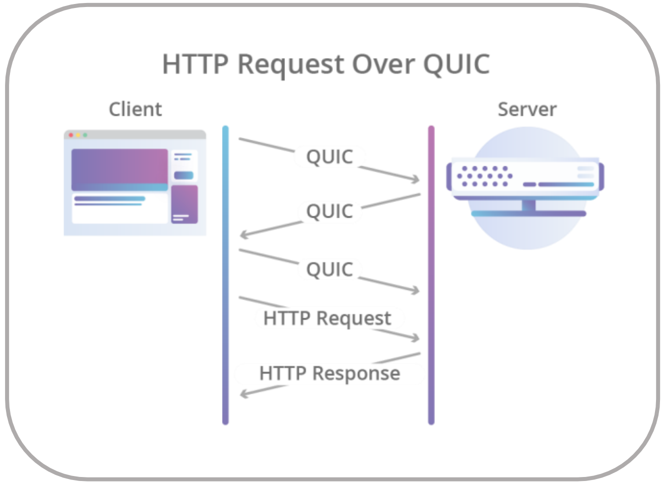
HTTP/2 over TCP

HTTP Over QUIC Network Stack

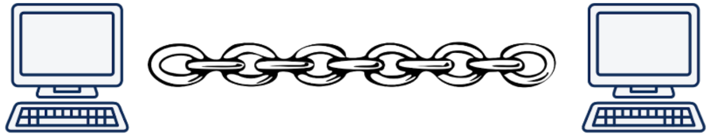


HTTP / 3

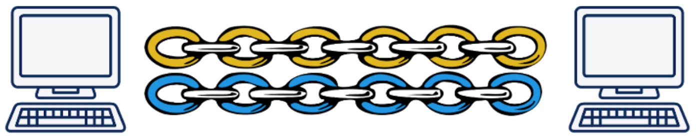
Workgroup:	QUIC
Internet-Draft:	draft-ietf-quic-http-latest
Published:	9 June 2020
Intended Status:	Standards Track
Expires:	11 December 2020
Author:	M. Bishop, Ed.
	Akamai



- **No - TCP head of line blocking**
  - **streams are independent to each other**
- **Faster handshake**
  - **Earlier data**
- **More encryption, always**
- **Over UDP (*Connection less, No resend, No flow control*)**



- Steam impact not connection impact



https://http3-explained.haxx.se/en/why-quic

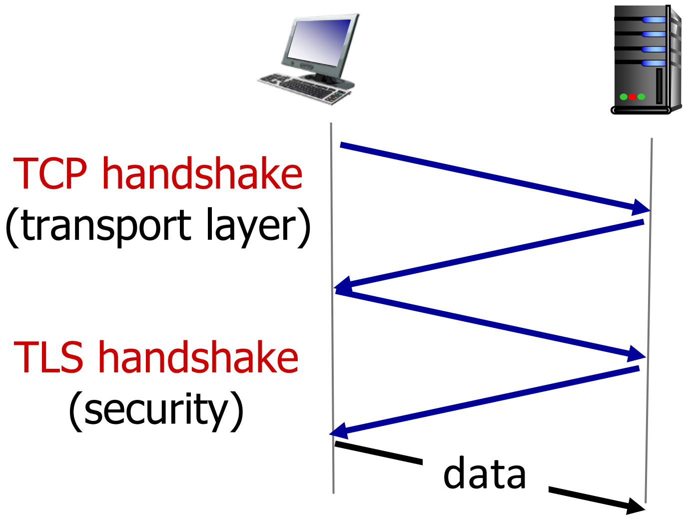
QUIC: Quick UDP Internet Connections

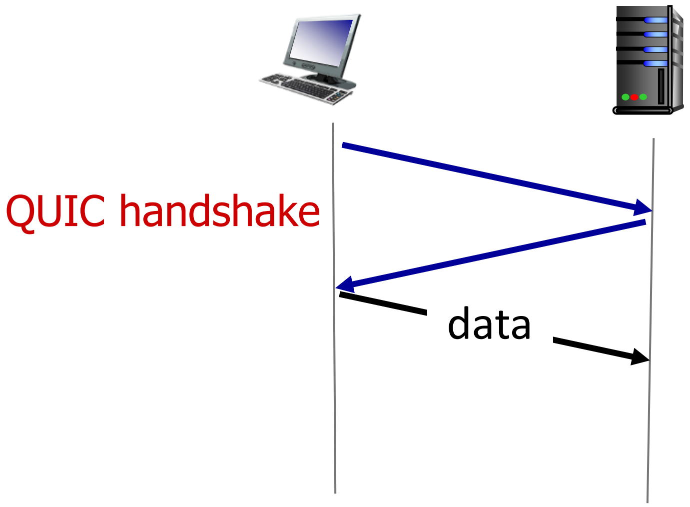
adopts approaches we've studied in this chapter for connection establishment, error control, congestion control

- **error and congestion control:** "Readers familiar with TCP's loss detection and congestion control will find algorithms here that parallel well-known TCP ones." [from QUIC specification]
- **connection establishment:** reliability, congestion control, authentication, encryption, state established in one RTT

- multiple application-level "streams" multiplexed over single QUIC connection
  - separate reliable data transfer, security
  - common congestion control

QUIC: Connection establishment





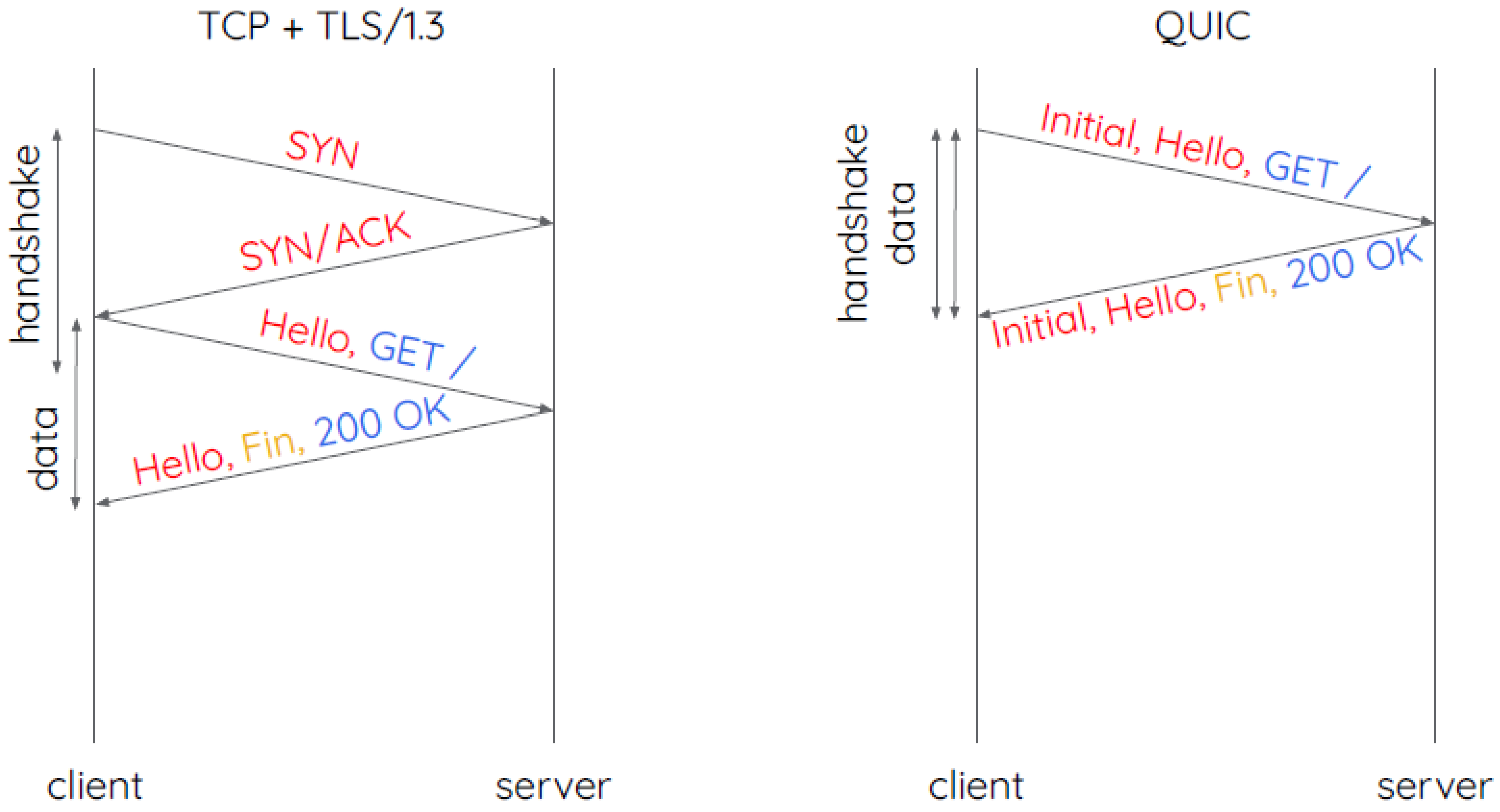
TCP (reliability, congestion control state) + TLS (authentication, crypto state)

- 2 serial handshakes

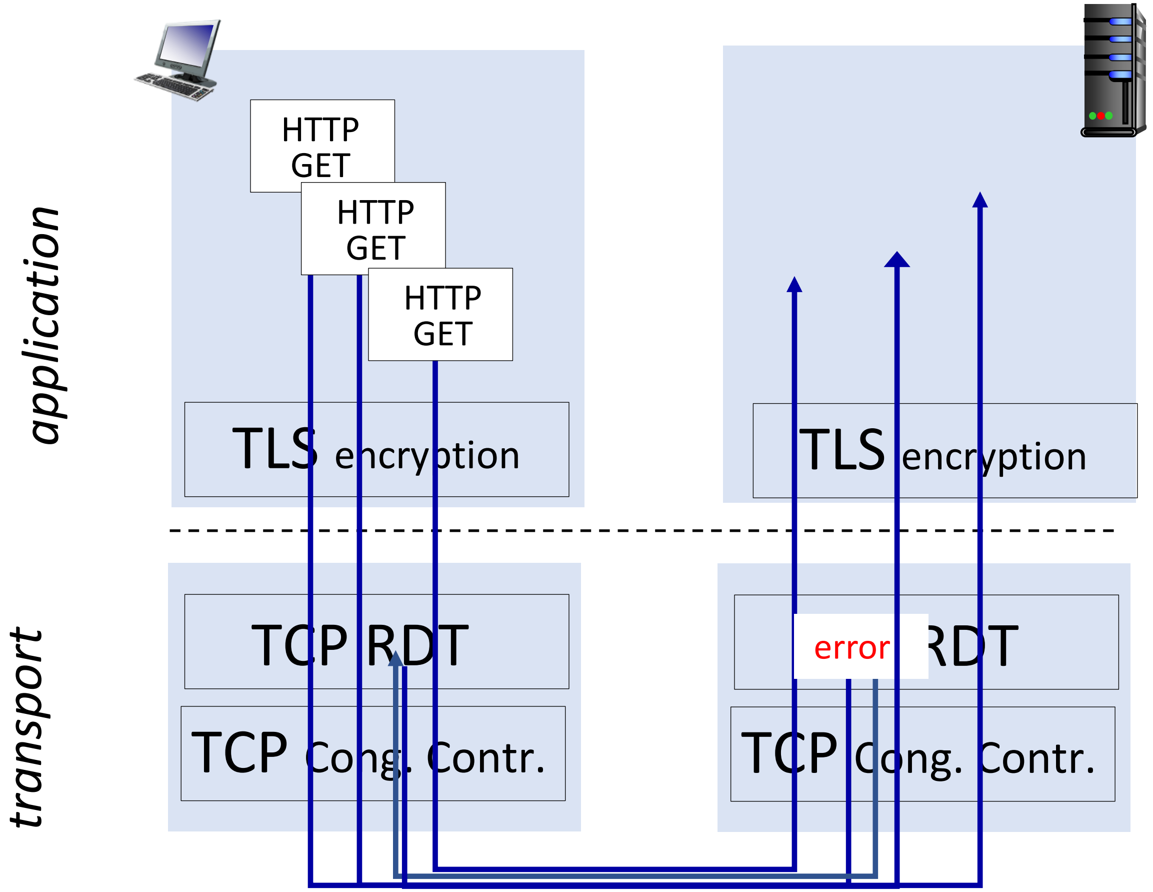
QUIC: reliability, congestion control, authentication, crypto state

- 1 handshake

Subsequent Connection to the same server



QUIC: streams: parallelism, no HOL blocking



(a) HTTP 1.1

QUIC Status

In May 2021, the IETF standardized QUIC in RFC 9000.

Implementations:
Apple, Facebook, Fastly, Firefox, F5, Google, Microsoft ...

Server deployments have been going on for a while
Akamai, Cloudflare, Facebook, Fastly, Google ...

Clients are at different stages of deployment
Chrome, Firefox, Edge, Safari
iOS, MacOS

Source: SIGCOMM 2020, QUIC Tutorial, https://conferences.sigcomm.org/sigcomm/2020/tutorial-quic.html

Network Layer

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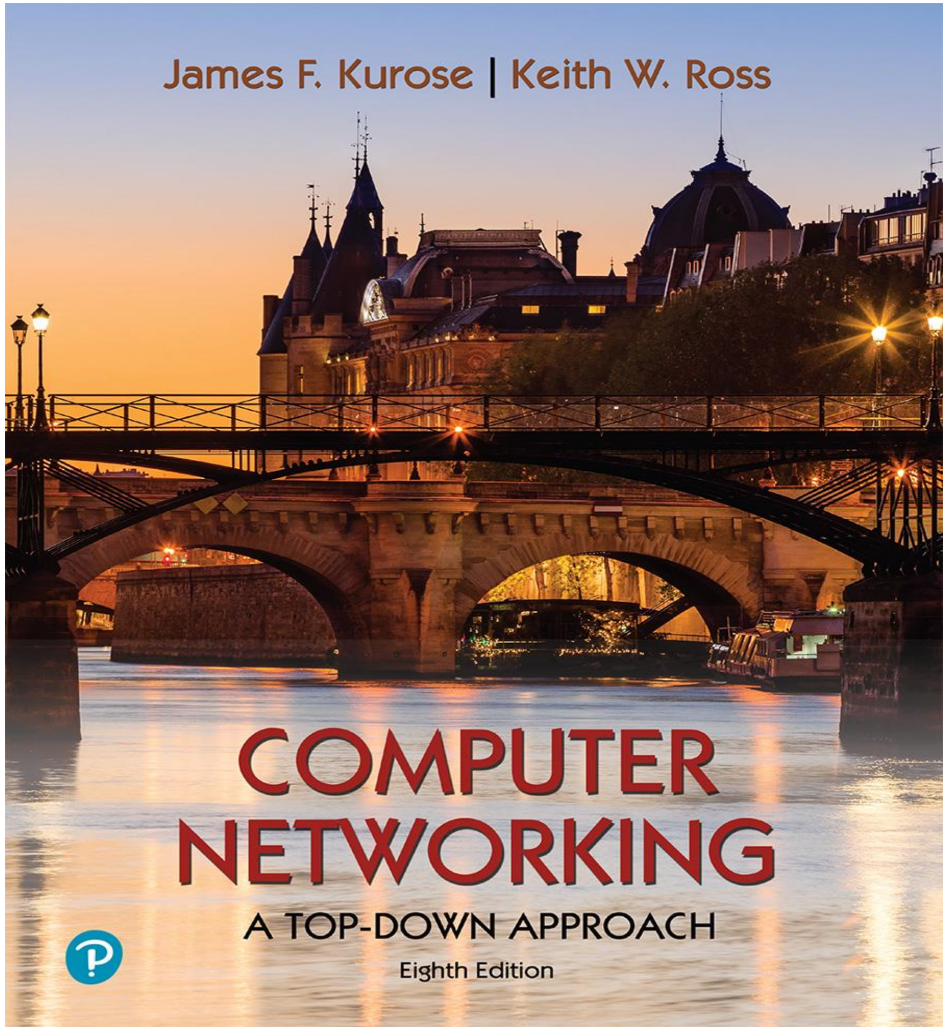
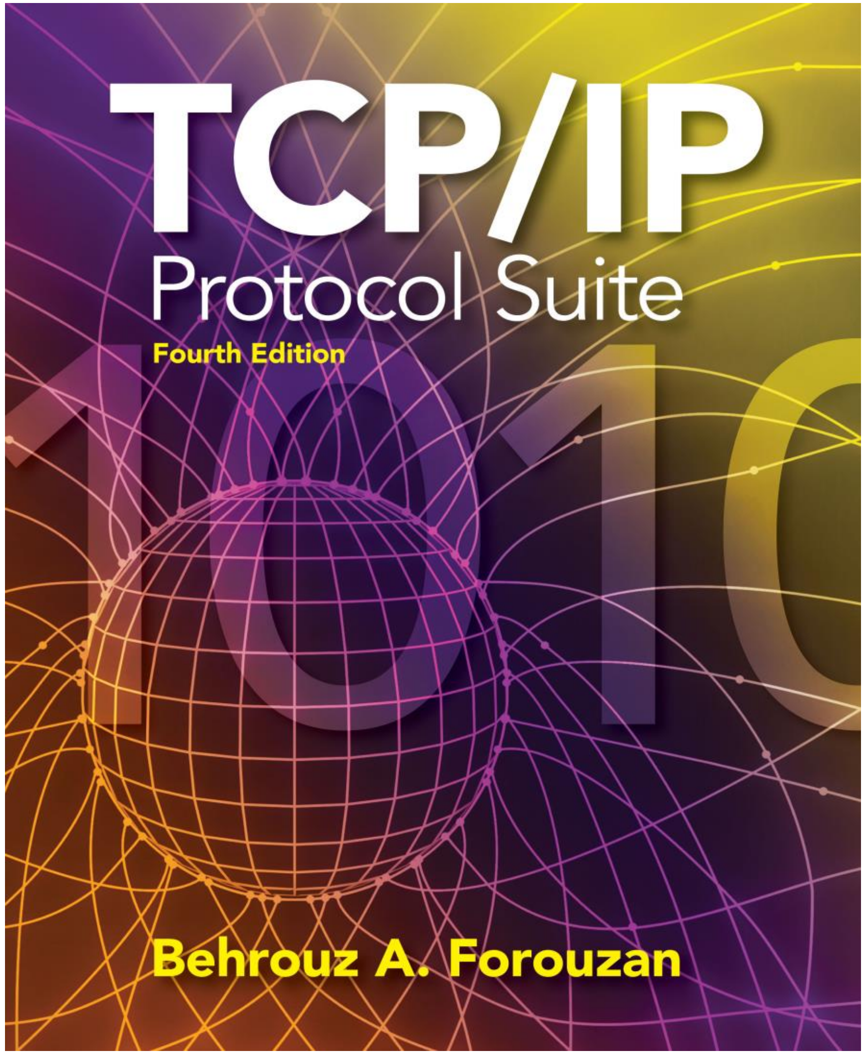
Sources



Computer Networking: A Top-Down Approach

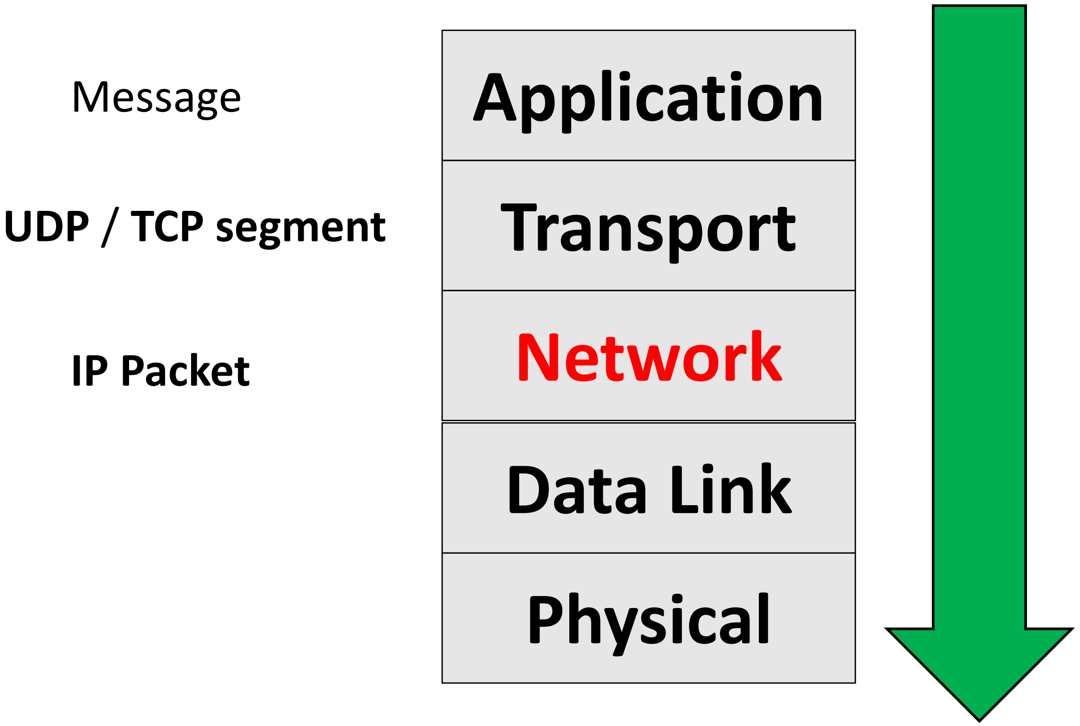


TCP/IP Protocol Suite

Outline

- Introduction to Network Layer (Overview)
- Network Layer Services
- IP Header Format - Fragmentation
- IP addressing
  - Subnetting, Supernetting
  - IP allocation - Dynamic Host Configuration Protocol (DHCP)
- Network Address Translation (NAT)
- Address Resolution Protocol (ARP)
- Internet Control Message Protocol (ICMP)
- Routing (RIP, OSPF, BGP)

Top Down Approach

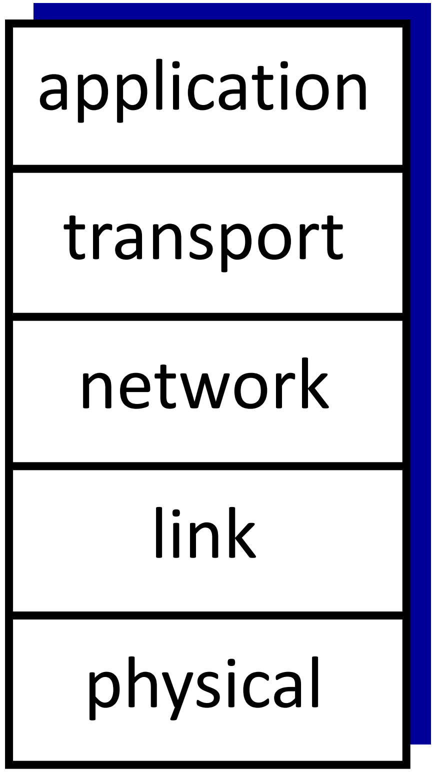


Internet protocol stack

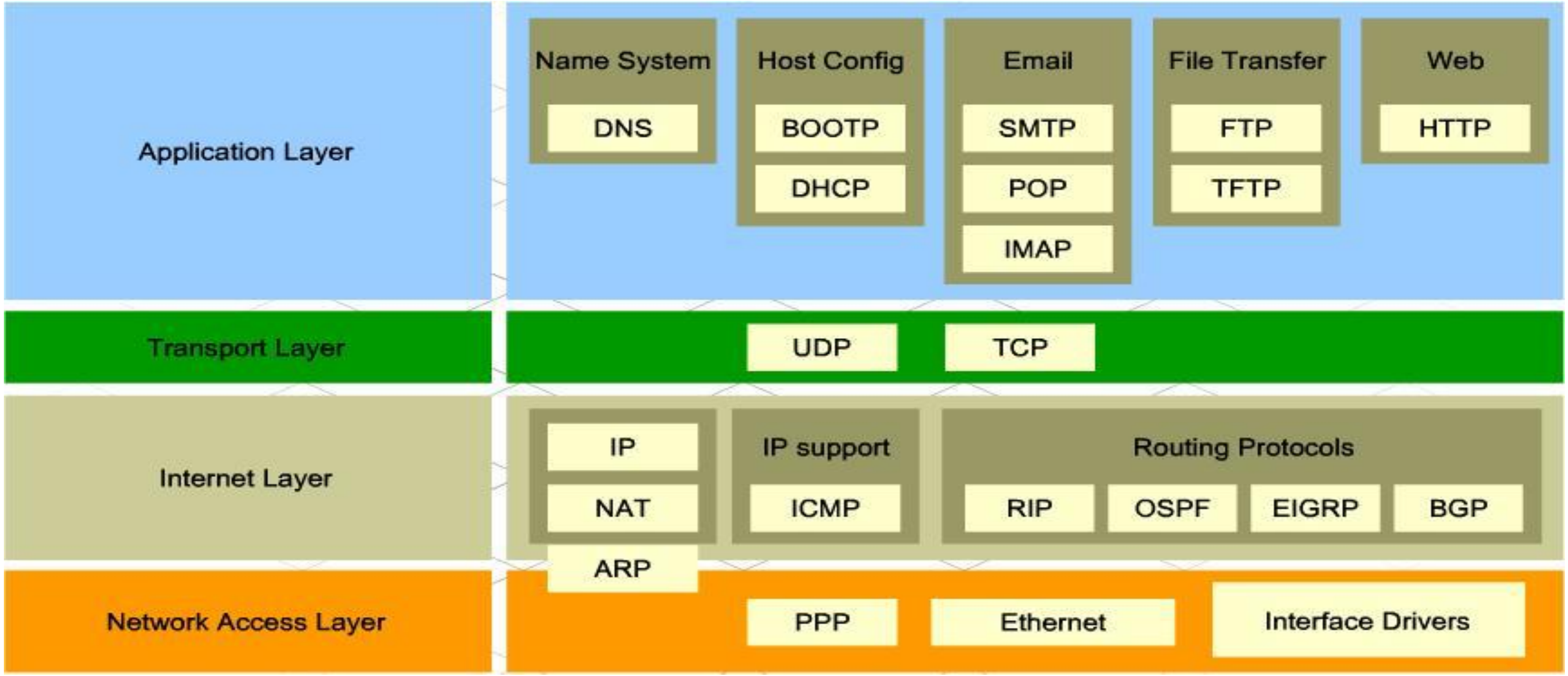
application: supporting network applications — HTTP, SMTP, IMAP, FTP

transport: process-process data transfer — TCP, UDP

network: routing of datagrams from source to destination — IP, routing protocols

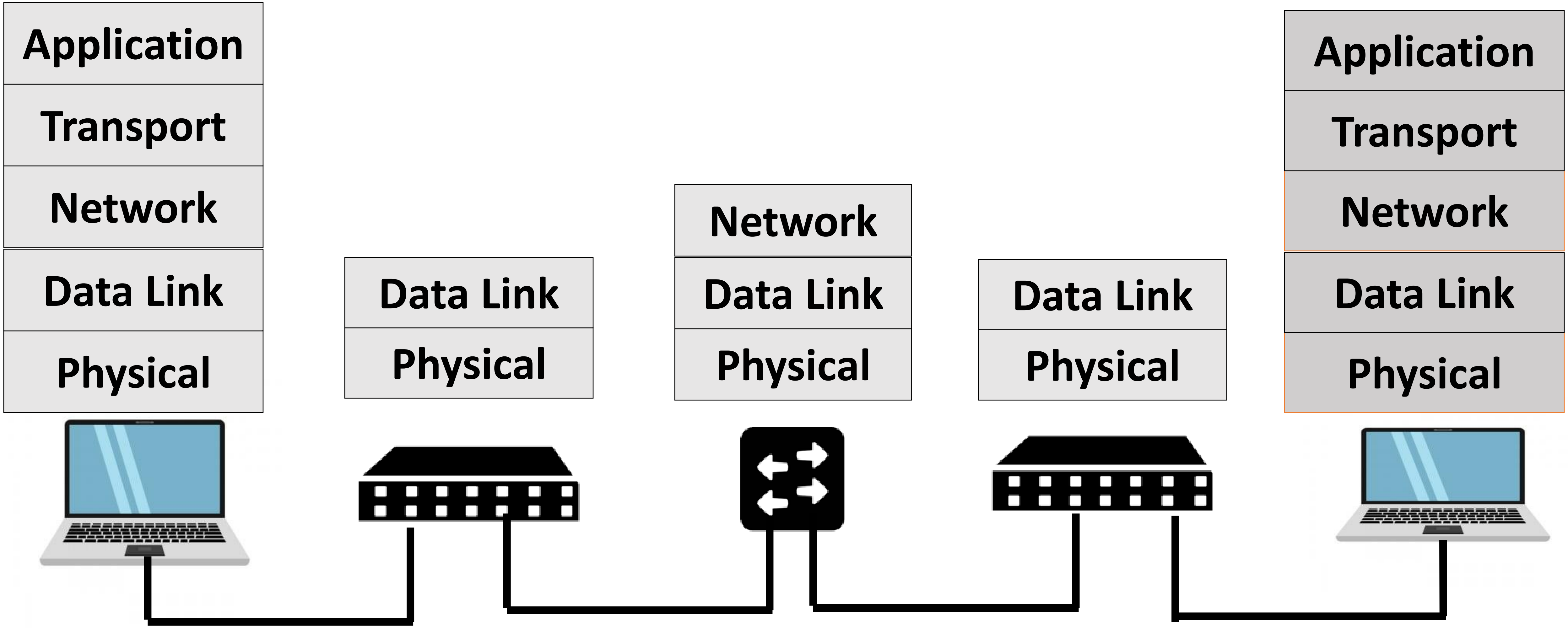


Protocols @ Different layers

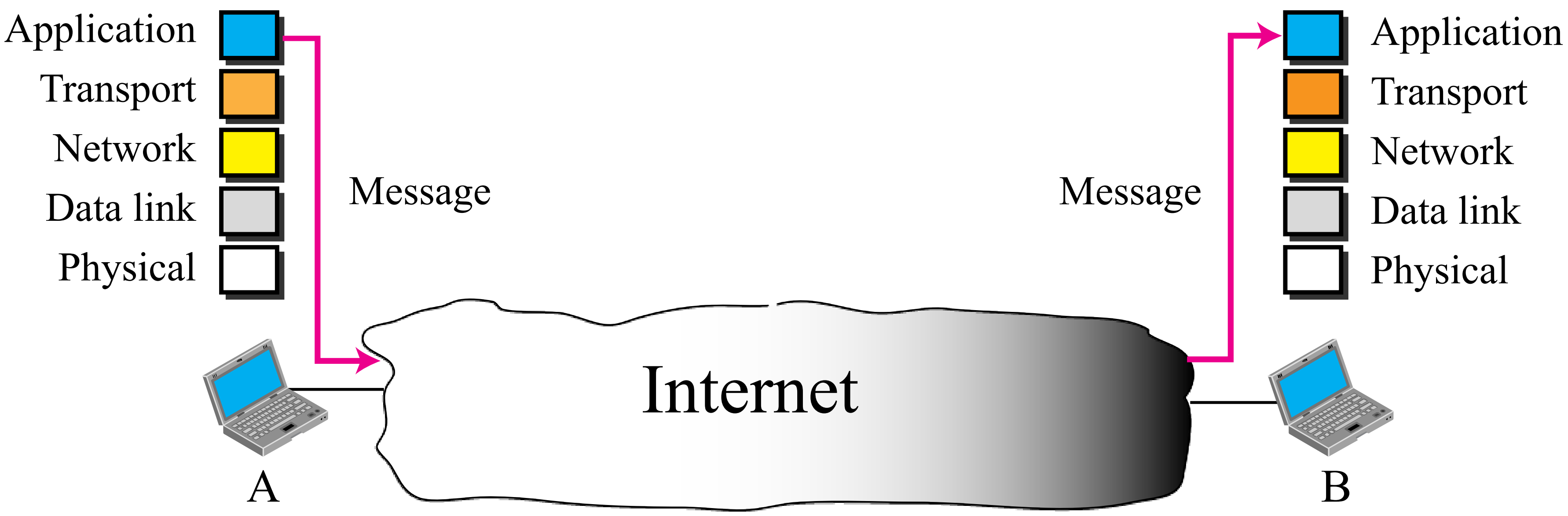


Source: http://walkwidnetwork.blogspot.com/2013/04/application-layer-internet-protocol.html

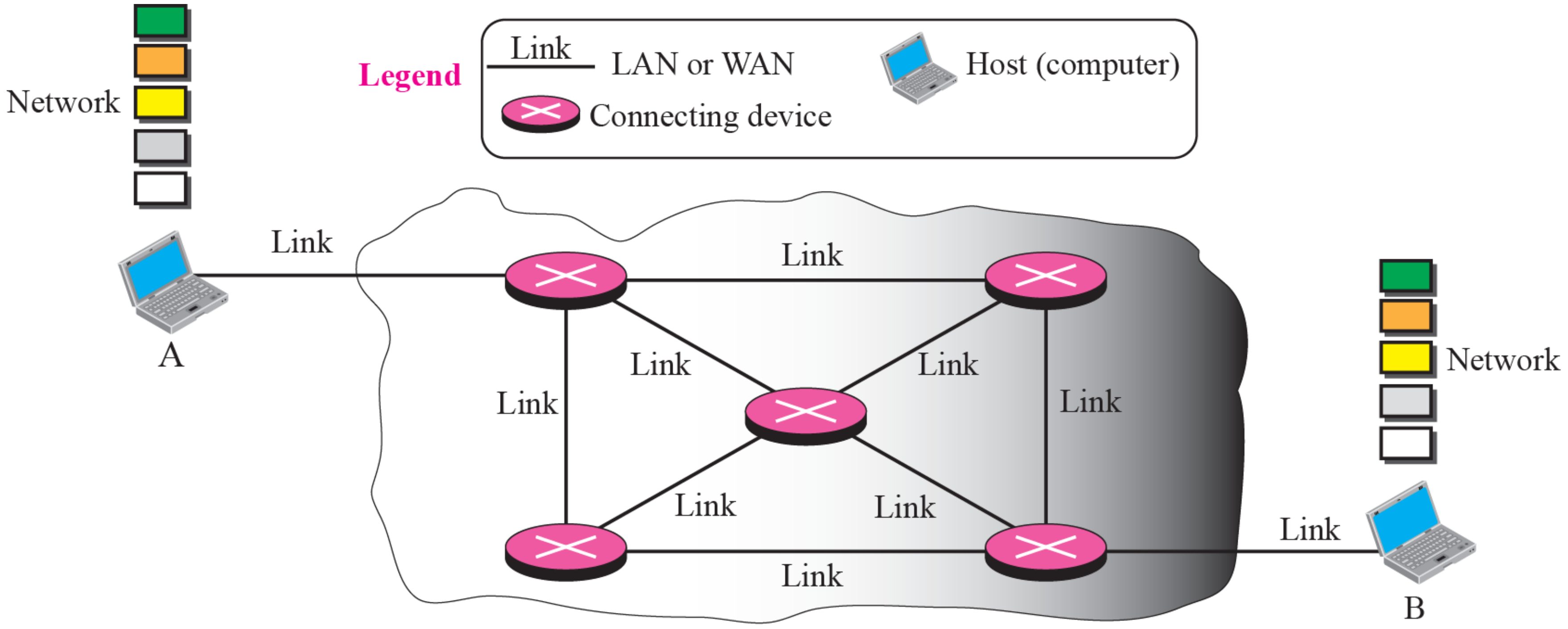
Communication between two remote Machine



Internet as a Black Box

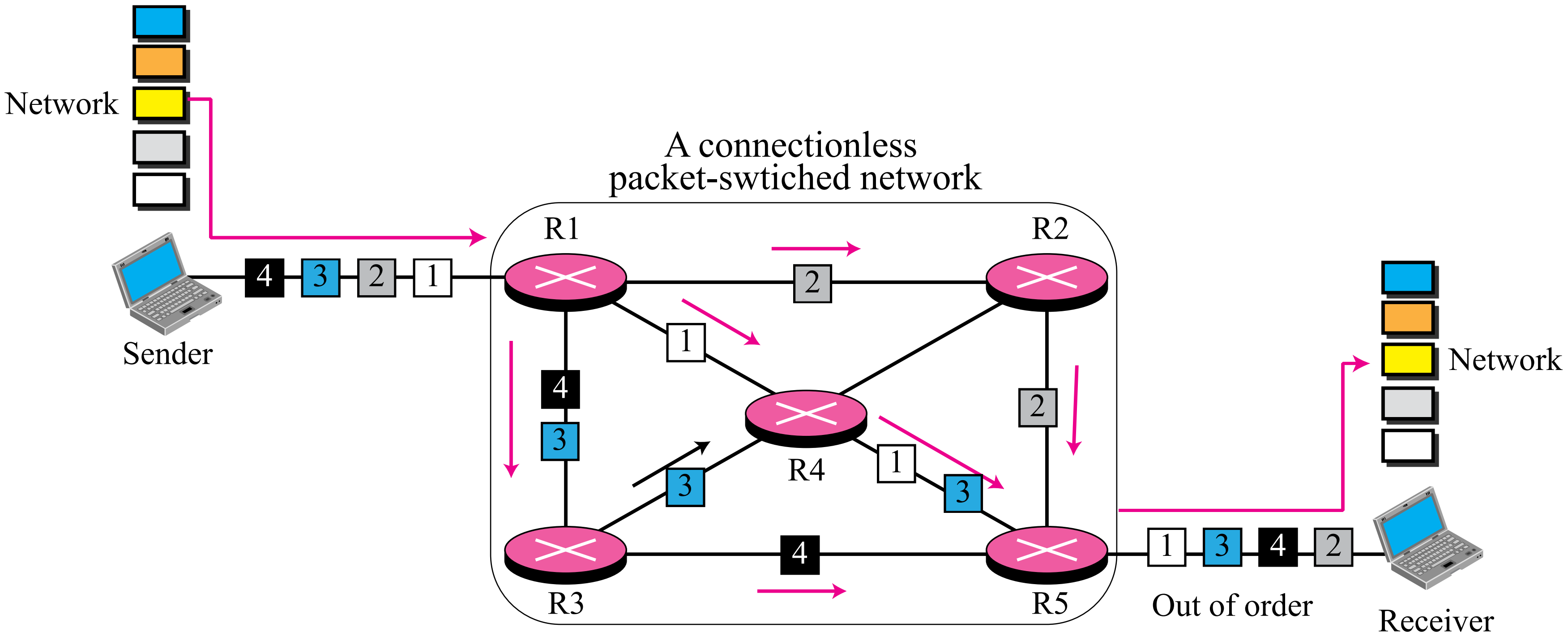


Internet as a combination of LANs and WANs connected together

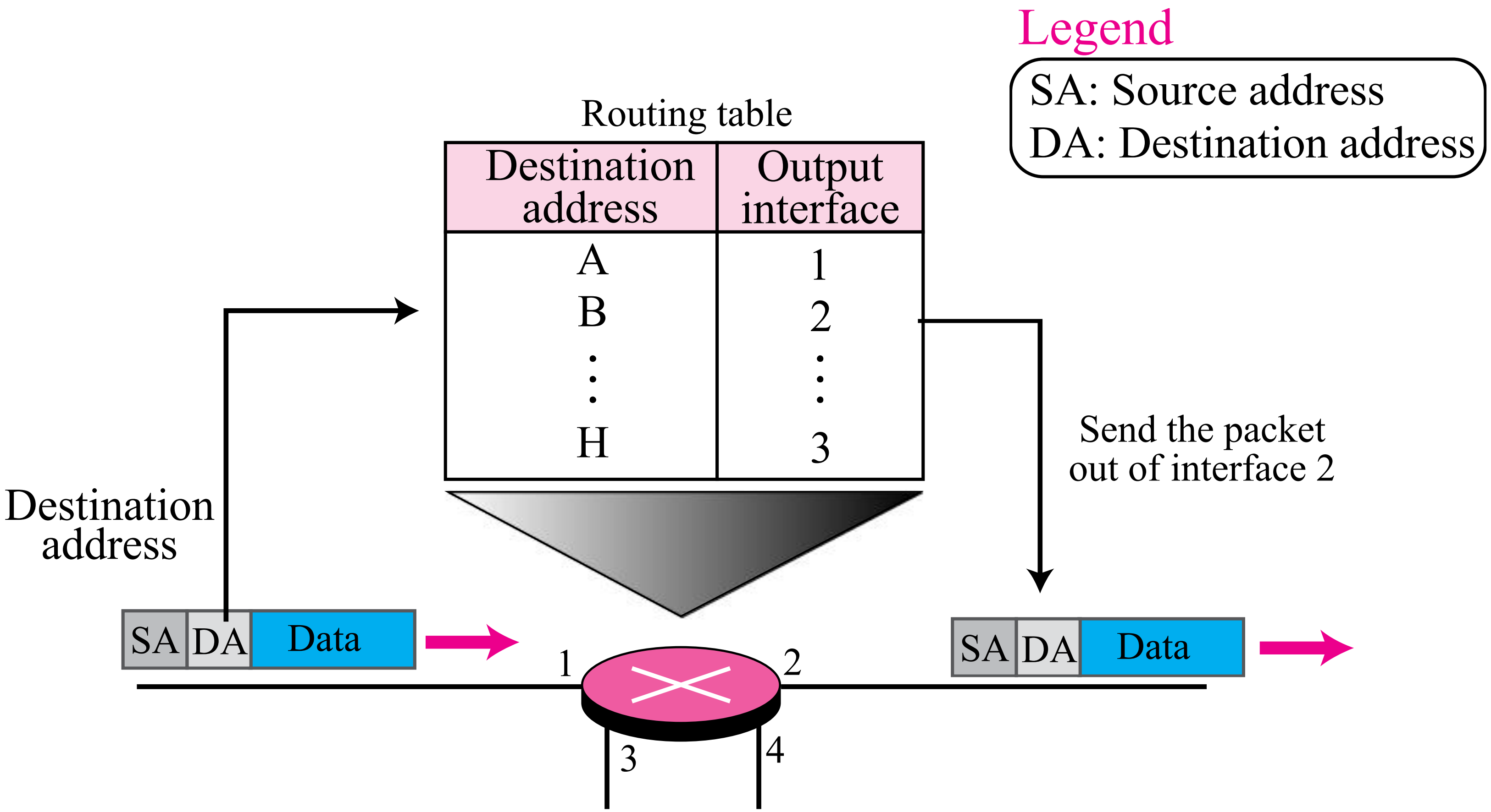


An internet as a combination of links and connecting devices

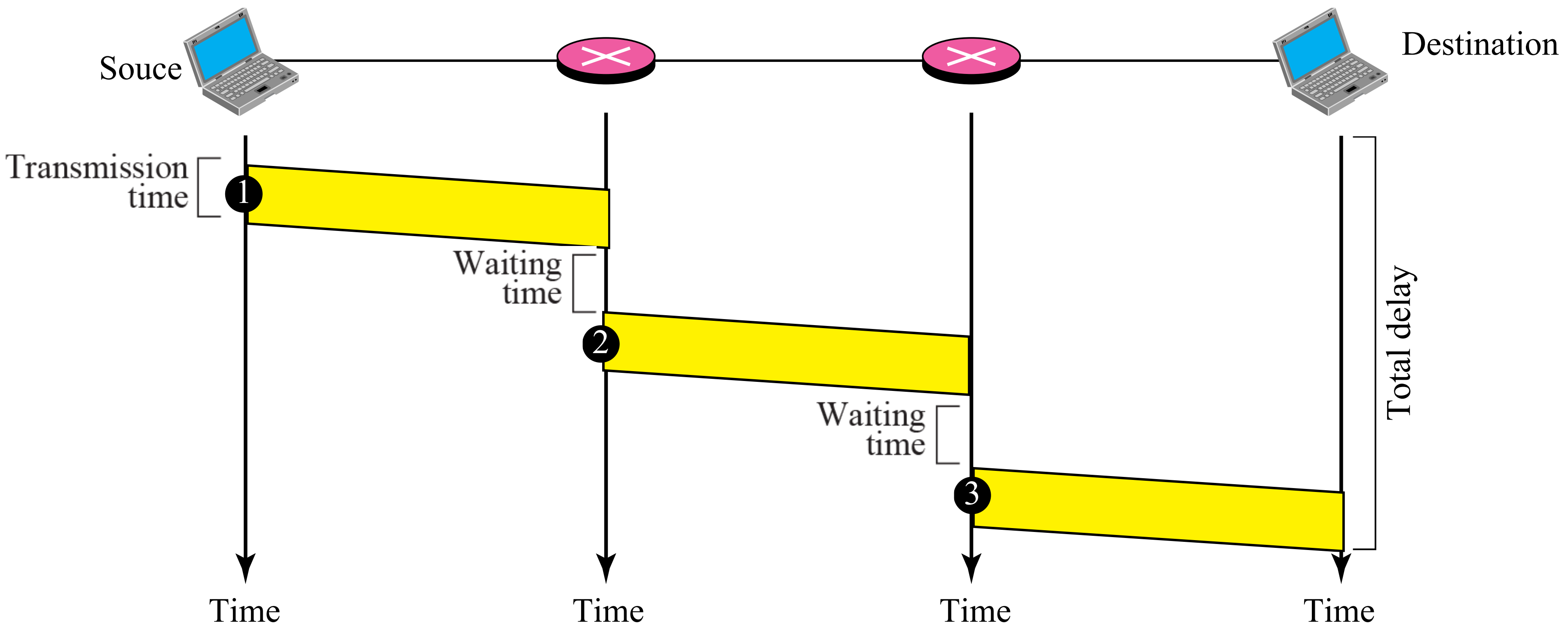
A connectionless packet-switched network



Forwarding process in a connectionless network



Delay in a connectionless network



Services provided at the source computer